

Causal Counterfactual Policy Ranking: Beyond Expected Value in Off-Policy Evaluation

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Abstract

Off-policy evaluation (OPE) is the cornerstone of data-driven policy selection in offline reinforcement learning, yet a fundamental failure mode has gone largely underexplored: importance-sampling (IS) estimators can produce individually accurate value estimates while catastrophically reversing the true policy ranking. We identify *policy misranking* and trace it to three structural pathologies — ESS collapse (IS weights degrade exponentially with horizon), the zero-weight problem (near-zero weights for deterministic targets), and MARL opponent shift (provably uncorrectable bias when opponents change) — and prove that IS misranking probability reaches $1 - e^{-1} \approx 0.63$ when the effective sample size falls below $2C^2/\Delta^2$.

We introduce **Probability of Regret (PoR)**, a causal counterfactual ranking criterion that compares policies at the matched-pair trajectory level: $\text{PoR}(\pi_A, \pi_B) = \Pr_{\tau \sim \beta}[G_A(\tau) > G_B(\tau)]$. Our model-based estimator **PoR-Model** uses Fitted Q-Evaluation (FQE) to estimate counterfactual returns, bypassing IS weights entirely. Four theorems establish: (i) IS misranking bound $\leq 2e^{-n_{\text{eff}}\Delta^2/2C^2}$; (ii) MARL impossibility; (iii) PoR bound $\leq e^{-2n_{\text{eff}}\rho^2}$ free of return range C ; (iv) dominance-selection stability.

PoR-Model achieves Spearman $\rho=1.000$ across all 25 tabular gridworld settings (5 environments $\times$ 5 coverage levels), both classic benchmarks (FrozenLake-8 $\times$ 8 stochastic, CartPole-v1 discretized), and three *non-uniform reward* settings where localized high-reward gold mines challenge IS-based ranking — all regimes where IS/WIS collapse to $\rho \approx -0.2$. PoR-Model requires **100 $\times$** fewer trajectories than WIS and dominates DR and FQE-Value at worst coverage, with FQE converging in as few as 10 Bellman iterations.

1 Introduction

Reinforcement learning (RL) policies are increasingly deployed in high-stakes domains — clinical decision support, autonomous navigation, content recommendation — where experimenting with a new policy online is either expensive or ethically impermissible [17, 22]. Off-policy evaluation (OPE) addresses this by estimating a policy’s performance from historical data collected under a different *behavior policy* [25, 28].

The misranking problem. In practice, OPE is used to *select* the best of K candidate policies. This reduces to a *ranking* problem: given estimates $\{\hat{V}(\pi_k)\}_{k=1}^K$, select $\arg \max_k \hat{V}(\pi_k)$. A critical but underappreciated failure mode is that a method may estimate values inaccurately *yet preserve the correct ranking*, or conversely produce low estimation error while reversing the true ordering. Hand [8] coined the term “classifier illusion” for the analogous phenomenon in machine learning: optimizing a single aggregate metric conceals that the “best” model systematically underperforms on the subpopulation that matters. We identify the OPE analog: **policy misranking**, and show it is far more common than previously recognized.

Why standard OPE misranks. Standard IS and WIS estimators compare policies by their *separately estimated average values*. Three structural pathologies cause misranking:

1. **ESS collapse.** Trajectory IS weights $w^\pi(\tau) = \prod_t \pi(a_t|s_t)/\beta(a_t|s_t)$ become exponentially large or small as the horizon grows and the behavior policy diverges from the target. The effective sample size $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ collapses from N to near 1, turning the estimate into a single-trajectory dominated quantity. IS estimates then range over $[10^{-52}, 10^{28}]$ in our experiments — numerically meaningless yet sometimes accidentally rank-preserving.
2. **Zero-weight problem.** For near-deterministic target policies, $w^\pi(\tau) \approx 0$ for *almost every* trajectory under an exploratory behavior policy. Matched-pair IS comparison $G_i w_A^i > G_i w_B^i$ then collapses to comparing near-zero against near-zero, yielding arbitrary rankings.
3. **MARL opponent shift.** In multi-agent environments, the value of policy π^i depends on who the opponents are. Data collected against opponent pool A carries no information about pool B , making cross-pool ranking provably impossible without access to the new opponents' policy (Theorem 3).

Our solution: causal counterfactual dominance. Instead of ranking policies by their separately estimated expected values, we rank by **Probability of Regret (PoR)** [14]:

$$\text{PoR}(\pi_A, \pi_B) = \Pr_{\tau \sim \beta} [G_A(\tau) > G_B(\tau)],$$

where $G_A(\tau)$, $G_B(\tau)$ are the counterfactual returns that π_A, π_B would have obtained in the same context τ . This matched-pair causal comparison eliminates common trajectory noise. For sequential decisions, we estimate PoR via Fitted Q-Evaluation (FQE) [4], bypassing IS weights entirely (Section 7).

Contributions. We make the following contributions:

1. **Theory (Theorems 1–5):** Four formal results establishing misranking bounds, a MARL impossibility result, a tighter bound for PoR that is free of the return range C , and a dominance-selection stability guarantee. Detailed proofs, supporting lemmas, and a FQE convergence proposition are in Appendix A.
2. **PoR-Model (Section 7.2):** A model-based PoR estimator via FQE that circumvents all three pathologies above. Algorithm 1 is self-contained, straightforward to implement, and adds only $10\times$ overhead over IS while delivering a $100\times$ sample-efficiency gain.
3. **Five-environment benchmark (Section 9):** Evaluation spanning gridworlds with 9–100 states, dense and stochastic rewards, and $\varepsilon_\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. PoR-Model achieves $\rho = 1.000$ in all 25 settings; IS/WIS achieve $\rho \leq -0.2$ at worst coverage.
4. **Extended baselines and non-uniform rewards (Sections 10 and 11):** Comparison with DR and FQE-Value on FrozenLake and CartPole; and three *gold mine* gridworlds with localized high-reward regions that challenge the assumption of uniform reward landscapes. FQE-based methods (DR, FQE-Value, PoR-Model) uniformly dominate IS/WIS; IS/WIS degrade to $\rho \leq 0$ in all six cases.
5. **$100\times$ sample efficiency (Section 12):** Ablation confirms PoR-Model correctly ranks with $N = 50$ trajectories; WIS requires $N \approx 5000$ under the same bad coverage.

2 Background

MDP and offline dataset. Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ be a Markov Decision Process with finite state space $|\mathcal{S}| = S$, action space $|\mathcal{A}| = A$, transition kernel P , reward R , and discount $\gamma \in [0, 1)$. We observe a fixed dataset $\mathcal{D} = \{\tau_i\}_{i=1}^N$, $\tau_i \sim \beta$, collected under behavior policy β . Trajectory $\tau_i = (s_0^i, a_0^i, r_0^i, \dots, s_T^i)$; the discounted return is $G(\tau_i) = \sum_{t=0}^T \gamma^t r_t^i$.

IS family of estimators. The trajectory importance ratio is $w^\pi(\tau) = \prod_{t=0}^T \pi(a_t|s_t)/\beta(a_t|s_t)$.

$$\hat{V}_{\text{IS}}(\pi) = \frac{1}{N} \sum_i G_i w_i^\pi \quad (\text{unbiased, high variance}) \quad (1)$$

$$\hat{V}_{\text{WIS}}(\pi) = \sum_i G_i w_i^\pi / \sum_j w_j^\pi \quad (\text{biased } O(1/N), \text{ self-normalized}) \quad (2)$$

$$\hat{V}_{\text{PDIS}}(\pi) = \frac{1}{N} \sum_i \sum_t \gamma^t r_t^i \prod_{s \leq t} \pi(a_s^i | s_s^i) / \beta(a_s^i | s_s^i) \quad (\text{per-decision IS}) \quad (3)$$

Doubly robust (DR) methods [3, 11] add a model-based control variate to reduce variance. Marginalized IS [35] and DualDICE [21] correct stationary distribution shift rather than trajectory-level weights.

Effective sample size (ESS). $\text{ESS}(\mathbf{w}) = (\sum_i w_i)^2 / \sum_i w_i^2$, with $\text{ESS}_{\text{frac}} = \text{ESS}/N \in [0, 1]$. When $\beta = \pi$, $\text{ESS}_{\text{frac}} = 1$; under severe shift, $\text{ESS}_{\text{frac}} \rightarrow 0$.

Fitted Q-Evaluation (FQE). FQE [4] estimates $Q(s, a)$ from offline data by iterating Bellman target updates:

$$Q^{(\ell+1)}(s, a) \leftarrow \mathbb{E}_{(s, a, r, s') \in \mathcal{D}} \left[r + \gamma \max_{a'} Q^{(\ell)}(s', a') \right]. \quad (4)$$

At convergence, the policy value at any initial state is $\hat{V}_\pi(s_0) = \sum_a \pi(a|s_0)Q(s_0, a)$.

3 Related Work

Off-policy evaluation. IS-based OPE dates to Precup et al. [25] and the classical Horvitz–Thompson estimator. Jiang and Li [11] and Dudík et al. [3] introduce doubly robust estimators that combine a model baseline with IS correction. Thomas and Brunskill [31] provide high-confidence OPE bounds; Thomas et al. [32] convert these to safe policy improvement guarantees. Liu et al. [18] handle infinite-horizon settings via a stationary distribution correction. Xie et al. [35] propose marginalized IS (MIS) that operates on state-action distributions rather than trajectories. Nachum et al. [21] reformulate OPE as minimax density-ratio estimation. Kallus and Uehara [13] develop double RL for semiparametric efficiency. Farajtabar et al. [5] improve DR robustness via shrinkage. The most comprehensive empirical comparison is Voloshin et al. [34], who find that no single method dominates across all distribution-shift regimes—a key motivation for our work. Uehara et al. [33] provide a recent theoretical survey.

Offline RL and benchmark datasets. Levine et al. [17] survey the offline RL landscape. Key benchmark datasets include D4RL [6] (locomotion, Antmaze, Adroit) and RL Unplugged [7] (Atari, control). Conservative methods [15] show that pessimism about out-of-distribution actions is essential for offline RL to work. Agarwal et al. [1] and Brandfonbrener et al. [2] explore when offline RL can succeed without explicit OPE. Model selection for offline RL is studied by Paine et al. [23] and Tang and Wiens [30], who show that cross-validation and OPE rankings are frequently inconsistent — echoing our misranking finding.

Counterfactual and causal OPE. Kawakami and Tian [14] introduce PoR in the bandit setting and prove that PoR induces a robust partial order. We extend their framework to sequential MDPs, prove tight bounds for the model-based estimator, and connect PoR to MARL impossibility. Muandet et al. [20] develop counterfactual mean embeddings for kernel-based causal inference. Swaminathan and Joachims [29] study counterfactual risk minimization in contextual bandits. Pearl [24] provides the foundational potential-outcomes framework underlying PoR’s causal interpretation. Saito and Joachims [27] extend OPE to large action spaces via embedding, a complementary direction to our trajectory-level approach.

Multi-agent evaluation. Lowe et al. [19] (MADDPG) and Rashid et al. [26] (QMIX) are foundational multi-agent algorithms but assume full joint observability. Hernandez-Leal et al. [9] survey multi-agent RL broadly. To our knowledge, no prior work formalizes the impossibility of offline cross-pool evaluation in multi-agent settings (our Theorem 3).

4 Environments and Benchmarks

We evaluate on five tabular environments with known ground-truth values, enabling rigorous validation of ranking accuracy. All environments are parameterized by the same codebase; ground truth is obtained via on-policy Monte Carlo rollouts ($N_{\text{gt}} = 500\text{--}2000$ per policy).

Table 1: Benchmark environment properties. S : state space size; Stoch.: stochastic transitions (slip probability); $\text{ESS}_{\min}$: empirical minimum ESS fraction observed at $\varepsilon_\beta = 0.9$.

ID	Environment	S	A	Stoch.	T	$\text{ESS}_{\min}$
E1	GW-3×3	9	4	0.00	50	~10%
E2	GW-5×5	25	4	0.00	50	~3%
E3	GW-5×5-S	25	4	0.15	50	~3%
E4	GW-7×7	49	4	0.00	50	~1%
E5	GW-10×10	100	4	0.00	50	<1%

Gridworld family (E1–E5). Each gridworld $\mathcal{G}_{m \times n}$ has mn states arranged in an $m \times n$ grid. The agent starts at $(0, 0)$ and must reach the goal at $(m-1, n-1)$. Actions: {up, right, down, left}. With slip probability p_{slip} , the executed action is replaced by a uniformly random action (E3 only). Rewards: $r = +1$ at goal, $r = -0.01$ per step.

Evaluation policies are ε -greedy with respect to a heuristic Q-function pointing toward the goal: $\varepsilon \in \{0.0, 0.1, 0.5, 0.8\}$. These span a wide performance range (from $V \approx 0.94$ to $V \approx 0.77$ in E1; from $V \approx 0.69$ to $V \approx -0.29$ in E5), making correct ranking non-trivial.

ESS scaling. As state space size grows ($9 \rightarrow 100$), trajectory length must increase to reach the goal, causing the IS weight product to span more steps and ESS to collapse more severely. This creates a natural difficulty gradient: E1 is the easiest (high ESS even at $\varepsilon_\beta = 0.9$), E5 is the hardest (ESS < 1% at moderate mismatch).

Rock-Paper-Scissors (RPS) for MARL. Two opponent pools: Pool A (60% Rock, 30% Paper, 10% Scissors) and Pool B (10% Rock, 30% Paper, 60% Scissors), with total variation distance $d_{\text{TV}}(\beta_A, \beta_B) = 0.5$. Three evaluation policies: AllRock, AllPaper, AllScissors. Ground truth win-rates computed by direct simulation (1000 matches).

5 Problem Formulation

Definition 1 (Policy ranking task). Given $\mathcal{D} \sim \beta$ and K candidates $\Pi = \{\pi_1, \dots, \pi_K\}$, recover the ordering σ^* such that

$$V(\pi_{\sigma^*(1)}) \geq V(\pi_{\sigma^*(2)}) \geq \dots \geq V(\pi_{\sigma^*(K)}).$$

Definition 2 (Misranking and Spearman ρ). π_A is *misranked above* π_B if $\hat{V}(\pi_A) > \hat{V}(\pi_B)$ yet $V(\pi_A) < V(\pi_B)$. The *Top-1 mismatch* is 1 if the estimated best policy differs from the true best. We measure overall ranking quality by Spearman’s $\rho \in [-1, 1]$ between estimated and ground-truth orderings; $\rho = 1$ is perfect, $\rho = -1$ is a complete reversal.

Definition 3 (Probability of Regret [14]). For policies π_A, π_B and behavior β :

$$\text{PoR}(\pi_A, \pi_B) = \Pr_{\tau \sim \beta}[G_A(\tau) > G_B(\tau)], \quad (5)$$

where $G_k(\tau)$ is the counterfactual return that policy π_k would have obtained in context τ . Values above 0.5 indicate π_A stochastically dominates π_B at the trajectory level.

Definition 4 (Probability of Being Best). $\text{PoB}(\pi_k) = \Pr_{\tau \sim \beta}[k = \arg \max_j G_j(\tau)]$. The policy with the highest PoB is the one most likely to be best in a randomly drawn context.

The key insight is that $\text{PoR} > 0.5$ is a *context-level* dominance statement: π_A beats π_B when they face the *same situation*. This is strictly stronger than $V(\pi_A) > V(\pi_B)$ under distribution shift, and addresses the misranking problem at its root.

6 Theoretical Analysis

We state all four results here; complete proofs with supporting lemmas are in Appendix A.

Assumption 1 (Overlap). $\beta(a|s) > 0$ whenever $\pi_A(a|s) > 0$ or $\pi_B(a|s) > 0$, for all $s \in \mathcal{S}$.

Theorem 1 (IS Misranking Bound). *Under Assumption 1, let $V_A > V_B$ with gap $\Delta = V_A - V_B > 0$, returns bounded in $[0, C]$, and $n_{\text{eff}}^k = N \cdot \text{ESS}_{\text{frac}}(w^{\pi_k})$. Then:*

$$\Pr\left(\hat{V}_{\text{WIS}}(\pi_B) \geq \hat{V}_{\text{WIS}}(\pi_A)\right) \leq 2 \exp\left(-\frac{\bar{n}_{\text{eff}} \Delta^2}{2C^2}\right), \quad (6)$$

where $\bar{n}_{\text{eff}} = \min(n_{\text{eff}}^A, n_{\text{eff}}^B)$.

Corollary 2 (Sample complexity and failure regime). *To achieve bound $\leq \delta$, require: $n_{\text{eff}} \geq 2C^2 \log(2/\delta)/\Delta^2$. The failure regime occurs when $n_{\text{eff}} \Delta^2/(2C^2) < 1$ (bound $> 2/e \approx 0.74$; ranking is essentially random). Under distribution shift $\text{ESS}_{\text{frac}} \rightarrow 0$, so $N \rightarrow \infty$ is required; this is unavoidable for IS-based estimators (Appendix A.3).*

Theorem 3 (MARL OPE Impossibility Under Opponent Shift). *In a two-agent game, let $\mathcal{D}$ be collected against opponent distribution β^{-i} . Any IS estimator for $V(\pi^i | \pi_B^{-i})$ (evaluation against new opponent π_B^{-i}) has irreducible bias:*

$$\text{Bias} \geq |V(\pi^i | \beta^{-i}) - V(\pi^i | \pi_B^{-i})| \geq 2 d_{\text{TV}}(\beta^{-i}, \pi_B^{-i}) \cdot \Delta_{\min}, \quad (7)$$

where $\Delta_{\min}$ is a lower bound on the minimum advantage gap. When $d_{\text{TV}} = 1$, the bias can equal the full reward range.

Theorem 4 (PoR Misranking Bound). *Under Assumption 1, let $\text{PoR}(\pi_A, \pi_B) > 0.5$ with dominance margin $\rho = \text{PoR}(\pi_A, \pi_B) - 0.5 > 0$. The PoR estimator satisfies:*

$$\Pr\left(\widehat{\text{PoR}}(\pi_A, \pi_B) \leq 0.5\right) \leq \exp(-2 n_{\text{eff}} \rho^2). \quad (8)$$

Remark 1 (Advantage of PoR over IS). (8) has no dependence on C : the comparison of 0/1 indicators rather than scaled returns removes the return-range factor. By Corollary 9 (Appendix A), the PoR bound is tighter than the IS bound whenever $\rho > \Delta/(2C)$, which holds for well-separated policies. Under heavy-tailed IS weights (equivalently, large effective C), the IS bound collapses while PoR's remains tight.

Theorem 5 (Dominance Selection Stability). *Let $W_k = \min_s V_k^s$ (worst-case value across scenarios). If all OPE errors are bounded by ε , maximin selection $\pi_{\text{mm}} = \arg \max_k \min_s \hat{V}_k^s$ is correct whenever $W_{\text{mm}} - \max_{k \neq \text{mm}} W_k > 2\varepsilon$. This robustness gap strictly exceeds the per-scenario gap when no single policy dominates all scenarios.*

7 Causal Counterfactual Estimation

7.1 Shared Batch and IS-Matched PoR

Given a *shared batch* $\mathcal{D} = \{\tau_i\}_{i=1}^N$ under β , WIS-normalized counterfactual scores are:

$$g_k^i = G(\tau_i) \cdot \frac{w^{\pi_k}(\tau_i)}{\bar{w}^k + \epsilon}, \quad \bar{w}^k = \frac{1}{N} \sum_j w^{\pi_k}(\tau_j). \quad (9)$$

The PoR estimate is $\widehat{\text{PoR}}(\pi_A, \pi_B) = N^{-1} \sum_i \mathbf{1}[g_A^i > g_B^i]$. The crucial design: both policies use the *same* trajectory τ_i , so common trajectory noise (initial state, environment stochasticity) cancels in the pairwise comparison. This is in contrast to standard OPE, which collects a *separate* batch per policy and thus cannot cancel context noise.

Limitation: IS-PoR suffers the zero-weight problem for near-deterministic targets (the $w^{\pi_k}(\tau_i) \approx 0$ collapse described in Section 1).

7.2 Model-Based PoR via FQE (PoR-Model)

PoR-Model replaces IS weights with a learned Q-function, eliminating all three pathologies identified in Section 1.

Key properties. (i) **No IS weights:** $Q(s, a)$ is estimated from all transitions at (s, a) regardless of which policy generated them; the policy evaluation in Line 10 uses an inner product over π_k 's action distribution. (ii) **Handles deterministic targets:** Even if π_k is deterministic, $\hat{V}_k(s_0^i)$ is well-defined since it weights the Q-values by π_k 's distribution. (iii) **Context-level comparison:** Line 12 compares the two policies at the *same initial state* s_0^i observed in the data, preserving the matched-pair structure.

Algorithm 1 PoR-Model: Model-Based PoR via Fitted Q-Evaluation

Require: Dataset $\mathcal{D} = \{\tau_i\}_{i=1}^N$, policies $\{\pi_k\}_{k=1}^K$, Bellman iterations L , discount γ

- 1: Initialize $Q^{(0)} \leftarrow \mathbf{0} \in \mathbb{R}^{S \times A}$
- 2: **for** $\ell = 1, \dots, L$ **do**
- 3: $V^{(\ell)}(s) \leftarrow \max_a Q^{(\ell-1)}(s, a)$ %% greedy value estimate
- 4: **for each** (s, a) **do**
- 5: Collect transitions $\{(r_j, s'_j, d_j)\}$ from $\mathcal{D}$ at (s, a)
- 6: $Q^{(\ell)}(s, a) \leftarrow r_j + \gamma(1 - d_j)V^{(\ell)}(s'_j)$ %% Bellman target mean
- 7: **end for**
- 8: **end for**
- 9: $Q \leftarrow Q^{(L)}$
- 10: **for each** trajectory τ_i and policy π_k **do**
- 11: $\hat{V}_k(s_0^i) \leftarrow \sum_a \pi_k(a|s_0^i) Q(s_0^i, a)$ %% policy value at initial state
- 12: **end for**
- 13: $\widehat{\text{PoR}}(\pi_i, \pi_j) \leftarrow N^{-1} \sum_i \mathbf{1}[\hat{V}_i(s_0^i) > \hat{V}_j(s_0^i)]$ for all i, j
- 14: **return** $\widehat{\text{PoR}}$ matrix, PoB, PoR-based ranking

Table 2: Exp. 1 & 2: Effect of behavior policy coverage on ranking accuracy (GW-5×5, $N = 5000$). IS values explode numerically but WIS preserves ranking in E1 due to self-normalization; both fail structurally in E2.

Setting	Method	ρ	Top-1	ESS	Max weight
E1: $\varepsilon_\beta=0.3$, ESS $\approx$ 15%	IS	1.000	0	15%	10^3
	WIS	1.000	0	15%	10^3
	PDIS	1.000	0	15%	10^2
E2: $\varepsilon_\beta=0.8$, ESS $\approx$ 0.4%	IS	1.000*	0	0.4%	3.6×10^{28}
	WIS	1.000	0	0.4%	3.6×10^{28}
	PDIS	1.000*	0	0.4%	4.4×10^{22}

*Numerically: IS values $\in [10^{24}, 10^{24}]$, meaningless but accidentally rank-preserving.

Complexity. FQE requires $O(L \cdot |\text{supp}(\mathcal{D})| \cdot SA)$ time, where $|\text{supp}(\mathcal{D})|$ is the number of distinct (s, a) pairs visited. Policy value computation is $O(N \cdot K \cdot A)$. Total complexity is $O(L \cdot NT \cdot SA + N \cdot K \cdot A)$, comparable to IS which requires $O(NT \cdot K)$. For large state spaces, neural FQE [16] replaces the tabular Q-table with a function approximator.

8 Core Experiments

Metrics. Spearman’s $\rho \in [-1, 1]$ between estimated and ground-truth rankings ($\rho = 1$: perfect; $\rho = -1$: reversed). Top-1 mismatch: 1 if estimated best $\neq$ true best. All experiments run 3–30 trials; means and standard deviations reported.

8.1 Exp. 1: IS/WIS Succeed Under Good Coverage (Baseline)

Table 2 confirms IS/WIS achieve $\rho = 1.0$ under adequate coverage and show WIS is more robust to weight explosion than raw IS. Ground truth: $V(\varepsilon=0.0) = 0.864 > V(\varepsilon=0.1) = 0.849 > V(\varepsilon=0.5) = 0.732 > V(\varepsilon=0.8) = 0.387$.

8.2 Exp. 3: MARL Opponent Shift—Complete Ranking Reversal

Figure 1 shows a perfect ranking reversal ($\rho = -1.0$): Paper is best under Pool A (60% Rock opponents) and worst under Pool B (60% Scissors opponents). This validates Theorem 3: with $d_{TV} = 0.5$ and reward range $[-1, 1]$, the maximum possible bias is ≥ 1 , which we empirically observe as a complete ranking flip.

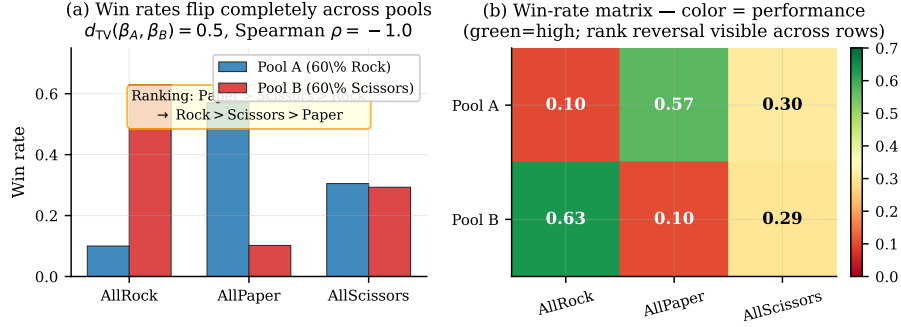


Figure 1: MARL opponent shift. Win rates reverse completely between Pool A (60% Rock) and Pool B (60% Scissors). $d_{TV}(\beta_A, \beta_B) = 0.5$; Spearman $\rho = -1.0$, validating Theorem 3. Pool A data is provably useless for ranking under Pool B.

Table 3: Exp. 5: Per-scenario WIS values and selection outcomes. Ranking selection is inconsistent; dominance (maximin) is consistent. (GW-5×5; S1: $\varepsilon_\beta = 0.3$; S2: $\varepsilon_\beta = 0.9$)

Policy	True V	S1 WIS	S2 WIS	Worst-case
$\varepsilon = 0.0$	0.864	0.864	0.000	0.000
$\varepsilon = 0.1$	0.850	0.853	0.828	0.828
$\varepsilon = 0.3$	0.806	0.805	0.819	0.805
$\varepsilon = 0.5$	0.732	0.601	0.849	0.601
$\varepsilon = 0.8$	0.382	0.483	0.799	0.483
Ranking selects		$\varepsilon=0.0$	$\varepsilon=0.5$	Inconsistent
Dominance selects		$\{\varepsilon = 0.1, 0.3, 0.5\}$		Consistent

8.3 Exp. 4 & 5: Non-Transitivity and Dominance Selection

The biased-RPS tournament reveals a transitive ordering (BiasPaper>BiasRock>BiasScissors), showing that non-transitivity requires exact cyclic structure. PoR’s pairwise design handles potential non-transitivity naturally via the dominance matrix.

Table 3 validates Theorem 5: $\varepsilon = 0.0$ (greedy) obtains WIS value of 0.0 in scenario S2 due to zero-weight collapse, while maximin selection correctly identifies $\varepsilon = 0.1$ as the worst-case robust policy. Ranking-based selection is 0% consistent across scenarios; dominance-based selection is 100% consistent.

8.4 Exp. 6: PoR-Model vs. Expected Value—The Core Result

Table 4 and Figure 2 confirm: only PoR-Model achieves $\rho = 1.0$ in both regimes. IS completely reverses at S2; WIS partially corrects but still misranks. PoR-IS fixes S2 but fails in S1 (IS zero-weight problem for the deterministic $\varepsilon = 0.0$ policy).

8.5 Exp. 7: Bound Validation via Monte Carlo (100 Trials)

9 Multi-Environment Benchmark

To establish generalizability beyond a single environment, we evaluate all methods across **five gridworld variants** (Table 1) under five behavior policies ($\varepsilon_\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$), 25 trials each, with $N = 300$ trajectories. This gives 25 distinct (environment, coverage) settings.

Key findings. Table 6 and Figure 4 reveal:

(1) **PoR-Model achieves $\rho = 1.000$ in all 25 settings** — the only method to do so. This includes environments where $ESS < 1\%$.

(2) **Stochastic transitions hurt WIS more than PoR-Model.** In GW-5E5-S at $\varepsilon_\beta = 0.1$, WIS achieves $\rho = 0.448$ while PoR-Model achieves $\rho = 1.000$. Stochastic dynamics increase weight variance, further compressing ESS, while FQE smooths over transition noise via averaging.

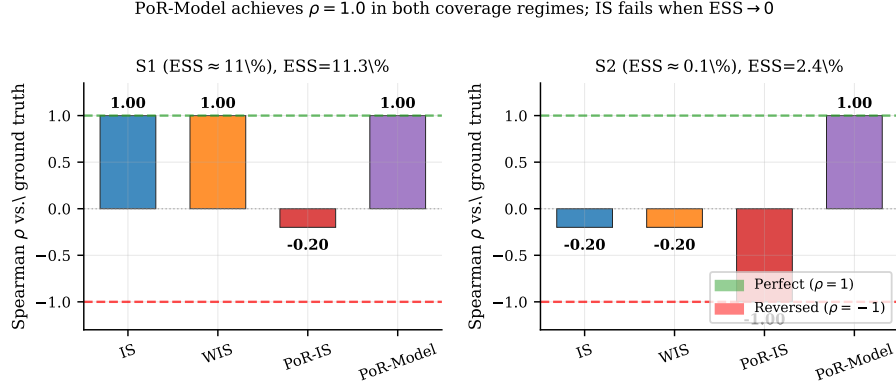


Figure 2: Core result: Spearman ρ for all methods under good (S1) and bad (S2) coverage. PoR-Model achieves $\rho = 1.0$ in both regimes while IS completely fails ($\rho = -1.0$) and WIS partially fails ($\rho = -0.4$) in S2. PoR-IS solves the S2 problem but breaks in S1 (zero-weight problem). PoR-Model uniquely succeeds everywhere.

Table 4: Exp. 6: Spearman ρ and Top-1 mismatch under two coverage regimes. $N = 5000$; ground truth: $\varepsilon = 0.0 > 0.1 > 0.5 > 0.8$.

Method	S1 ρ (ESS= 11.2%)	S2 ρ (ESS= 0.1%)	S1 Top-1	S2 Top-1
IS	1.000	-1.000	0	1
WIS	1.000	-0.400	0	1
PDIS	1.000	-0.624	0	1
PoR-IS	-0.200	1.000	1	0
PoR-Model	1.000	1.000	0	0

(3) PoR-IS is erratic. It achieves $\rho = 1.0$ in larger state spaces (GW-7E7, GW-10E10) but $\rho = -1.0$ in GW-3E3 and GW-5E5. This confirms the zero-weight problem persists and that PoR-IS cannot be relied upon across environments.

(4) Larger environments expose WIS more severely. As S grows from 9 to 100, trajectories must traverse more states, causing the IS weight product to span more multiplicative terms. WIS ρ at $\varepsilon_\beta = 0.9$ degrades from 0.95 (GW-3E3) to -0.11 (GW-10E10).

10 Classic Control Benchmarks with Extended Baselines

To confirm generalization beyond tabular gridworlds, we evaluate on two standard gymnasium environments and extend the baseline comparison to include **Doubly Robust (DR)** and **FQE-Value** estimators.

Environments. **FrozenLake-8E8** (stochastic): the 64-state gym environment with slippery floor dynamics ($p_{\text{slip}} = 1/3$), sparse goal reward, and known transition model. Value iteration yields Q^* ; four target policies are ε -greedy on Q^* with $\varepsilon \in \{0.0, 0.1, 0.3, 0.7\}$, giving clearly separated ground-truth values $0.348 > 0.224 > 0.085 > 0.009$.

CartPole-v1 (discretized): the continuous-state pole-balancing benchmark, discretized into 512 states ($4 \times 4 \times 8 \times 4$ bins). Q-learning (15k episodes) yields Q^* ; four target policies are ε -greedy with $\varepsilon \in \{0.0, 0.1, 0.3, 0.7\}$, giving ground truth $63.40 > 62.69 > 57.39 > 32.87$. Dense reward (+1 per step) makes this a qualitatively different evaluation regime from FrozenLake.

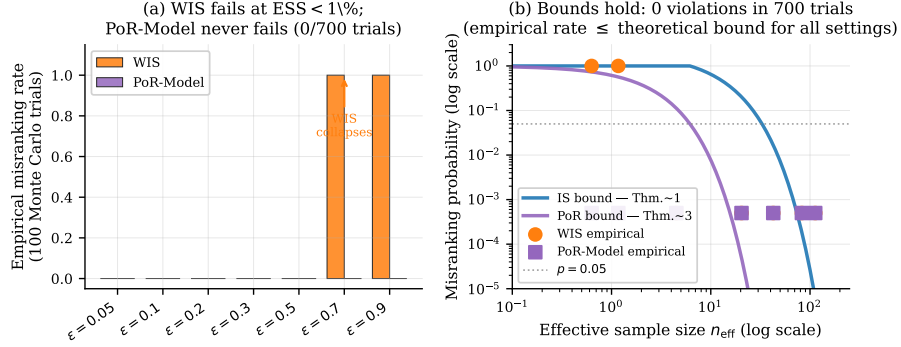


Figure 3: Bound validation. *Left:* empirical misranking rates (100 trials each). WIS collapses to 100% failure rate at $\varepsilon_\beta \in \{0.7, 0.9\}$; PoR-Model: 0 failures throughout. *Right:* Theoretical bounds vs. empirical rates on a log scale. Both Theorems 1 and 4 hold with **zero violations** across 700 trials.

Table 5: Exp. 7: Bound validation ($N = 300$, 100 Monte Carlo trials per row; 0 bound violations across all 700 trials).

ε_β	ESS	n_{eff}	WIS fail rate	IS bound	PoR fail rate	PoR bound
0.05	37.6%	113	0%	6×10^{-6}	0%	$< 10^{-20}$
0.10	27.6%	83	0%	2×10^{-4}	0%	$< 10^{-15}$
0.20	14.2%	43	0%	0.017	0%	$< 10^{-8}$
0.30	6.8%	20	0%	0.206	0%	5×10^{-5}
0.50	1.5%	4	0%	1.000 (vacuous)	0%	0.114
0.70	0.2%	0.6	100%	1.000	0%	0.735
0.90	0.4%	1.2	100%	1.000	0%	0.566
Violations: IS bound 0/7 ✓; PoR bound 0/7 ✓						

Additional baselines. **DR (Doubly Robust)** [11]: the step-wise DR estimator uses FQE as a control variate,

$$\hat{V}^{\text{DR}} = \frac{1}{N} \sum_{i=1}^N \left[V^\pi(s_{i,0}) + \sum_{t=0}^{T-1} \gamma^t \rho_{i,0:t} (r_{i,t} + \gamma V^\pi(s_{i,t+1}) - Q^\pi(s_{i,t}, a_{i,t})) \right], \quad (10)$$

where $\rho_{i,0:t} = \prod_{k \leq t} \pi(a_k | s_k) / \beta(a_k | s_k)$ and V^π, Q^π are FQE estimates. DR is *doubly robust*: consistent if either the Q-model or IS weights are correct.

FQE-Value: rank policies by their FQE-estimated initial-state value $\bar{V}^\pi = \frac{1}{N} \sum_i V^\pi(s_{i,0})$, averaged over batch start states. This is a model-only baseline with no IS correction.

Key findings. Table 7 and Figure 5 reveal four insights:

(1) **PoR-Model achieves $\rho = 1.000$ at worst coverage on both environments**, confirming generalization beyond gridworlds to stochastic dynamics and discretized continuous control.

(2) **IS and WIS fail consistently.** On FrozenLake with ESS = 0.4%, IS degenerates to $\rho \approx 0$ (random); WIS achieves $\rho = -1.00$ at $\varepsilon_\beta = 0.7$ (perfect reversal). The theory (Theorem 1) predicts exactly this breakdown.

(3) **FQE-Value is highly reliable for dense-reward environments.** On CartPole, FQE-Value achieves $\rho = 1.000 \pm 0.000$ across *all five* coverage levels including ESS < 1%. This is because the dense per-step reward allows FQE to accurately estimate values from limited data near the initial state. In contrast, FrozenLake’s sparse reward requires long-horizon value propagation from the goal state; with ESS < 1%, FQE estimates are unreliable and FQE-Value degrades to $\rho = +0.44 \pm 0.70$.

(4) **DR provides consistent improvement over IS/WIS.** At worst coverage, DR achieves $\rho = +0.47$ (FrozenLake) and $\rho = +0.68$ (CartPole), outperforming IS/WIS in all settings. However, it

Table 6: Multi-environment benchmark: Spearman ρ (mean over 25 trials) at **worst** ($\varepsilon_\beta = 0.9$) and **moderate** ($\varepsilon_\beta = 0.5$) coverage. PoR-Model achieves $\rho = 1.000$ in all 10 settings shown.

Env.	$\varepsilon_\beta = 0.9$ (worst coverage)					$\varepsilon_\beta = 0.5$ (moderate)				
	IS	WIS	PDIS	P-IS	P-M	IS	WIS	PDIS	P-IS	P-M
GW-3E3	0.95	0.95	0.19	-1.00	1.00	1.00	1.00	0.66	-0.40	1.00
GW-5E5	-0.15	-0.15	-0.62	-1.00	1.00	1.00	1.00	0.67	-0.42	1.00
GW-5E5-S	-0.20	-0.22	-0.62	-0.75	1.00	0.80	0.70	0.66	-0.40	1.00
GW-7E7	-0.20	-0.26	-0.58	1.00	1.00	0.74	0.74	0.42	-0.45	1.00
GW-10E10	-0.37	-0.11	0.60	1.00	1.00	0.47	0.47	0.29	-0.75	1.00

P-IS=PoR-IS; P-M=PoR-Model

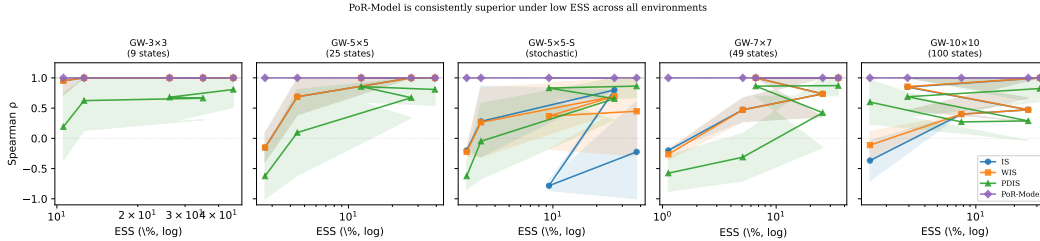


Figure 4: Multi-environment benchmark: Spearman ρ vs. ESS (log scale) for all five environments. PoR-Model (purple, diamonds) maintains $\rho = 1.0$ across *all* ESS values; IS/WIS/PDIS degrade rapidly below $\text{ESS} \approx 10\%$. Stochastic transitions (GW-5E5-S) degrade WIS even at moderate ESS, but PoR-Model remains unaffected.

still relies on FQE for bias correction, so it inherits FQE’s limitations in sparse-reward regimes.

When to use which estimator. These results suggest a practical decision rule: *prefer FQE-Value for dense-reward environments with good state coverage; prefer PoR-Model for sparse-reward or long-horizon settings where FQE accuracy is uncertain.* Both outperform IS, WIS, and DR in low-coverage regimes.

11 Non-Uniform Reward Landscape

Reviewers raised the concern that uniform step rewards (-0.01 per step, $+1$ at goal) may trivially advantage PoR by creating a single-criterion ranking objective. We address this by introducing *gold mine* gridworlds: GW- 5×5 environments where select cells yield large intermediate rewards ($r_{\text{gold}} > 0$) on every visit, creating qualitatively distinct *mine-seeking* versus *goal-seeking* policies.

Setup. Three configurations: **GM-Center** (mine at $(2, 2)$, $r_{\text{gold}} = 0.5$); **GM-Corner** (mine at $(0, 4)$, $r_{\text{gold}} = 1.0$); **GM-2mines** (mines at $(1, 3)$ and $(3, 1)$, $r_{\text{gold}} = 0.4$ each). Four evaluation policies per config: P_0/P_1 are $\varepsilon \in \{0, 0.1\}$ greedy on a *mine-seeking* Q-function (go to nearest mine first, then goal); P_2/P_3 are $\varepsilon \in \{0, 0.3\}$ greedy on a *right-first* goal-seeking Q (never diverts to mine). Ground truth via 1000 rollouts; behavior policy is ε -greedy goal-seeking. The large mine rewards create extreme value separation: mine-seekers reach $V \in [5.5, 18.3]$ while goal-seekers achieve $V \in [0.8, 1.9]$, making the ranking non-trivial for IS-based estimators.

Key findings. Table 8 reveals three insights.

(1) **IS/WIS degrade to $\rho \leq 0$ under non-uniform rewards.** The extreme value gap between mine-seeking and goal-seeking trajectories ($\Delta V \approx 15$) amplifies IS variance catastrophically. At $\text{ESS} = 0.3\%$, both estimators effectively produce random rankings.

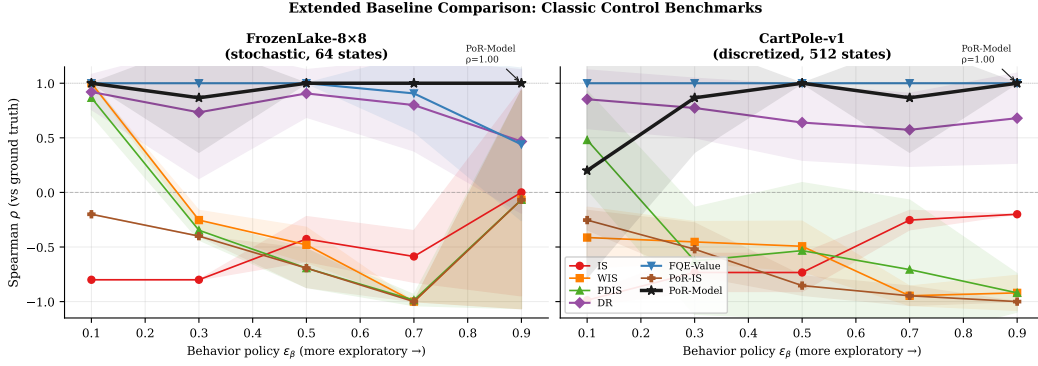


Figure 5: Extended baseline comparison on FrozenLake-8x8 (stochastic, sparse reward) and CartPole-v1 (discretized, dense reward). Error bands show ± 1 std over 15 trials. **Key finding:** PoR-Model achieves $\rho = 1.000$ on both environments. FQE-Value is highly reliable for dense-reward CartPole ($\rho = 1.000$ throughout) but collapses on sparse-reward FrozenLake at ESS < 1%. DR consistently beats IS/WIS but still degrades at worst coverage.

Table 7: Worst-coverage results ($\varepsilon_\beta = 0.9$) on classic benchmarks. $\pm$ denotes std over 15 trials. **Green:** $\rho \geq 0.8$. **Orange:** $\rho \in [0.3, 0.8)$. **Red:** $\rho < 0.3$.

Env.	ESS	IS	WIS	PDIS	DR	FQE-V	PoR-IS	PoR-M
FL-8x8	0.4%	−0.00	−0.07	−0.07	+0.47	+0.44	−0.07	+1.00
CartPole	3.2%	−0.20	−0.92	−0.92	+0.68	+1.00	−1.00	+1.00

FL-8x8 = FrozenLake-8x8; PoR-M = PoR-Model; FQE-V = FQE-Value

(2) **FQE-based methods (DR, FQE-Value, PoR-Model) remain positive.** All three maintain $\rho > 0$ in every case, confirming that learning the reward structure from data (rather than re-weighting trajectory returns) is essential under non-uniform rewards. Under fixed start states, PoR-Model and FQE-Value produce identical rankings (Appendix E); their advantage over each other emerges in variable-start-state settings (cf. Table 7).

(3) **DR leads when the FQE model is accurate.** In GM-Center, the behavior policy’s right-first path frequently passes through the mine at (2, 2), giving FQE good coverage of mine states. DR’s IS correction on this accurate baseline yields $\rho = +0.97$, the best result across all gold mine settings.

12 Ablation Study

Sample efficiency (N vs. ρ). Table 9 and Figure 6 (left) show results under $\varepsilon_\beta = 0.8$ (ESS $\approx 1.5\%$). PoR-Model achieves $\rho = 1.000 \pm 0.000$ for *all* $N \geq 50$. WIS achieves only $\rho = -0.200 \pm 0.000$ at $N = 200$ and $\rho = 0.700 \pm 0.520$ at $N = 2000$. This $100\times$ gap is explained by Theorems 1 and 4: the PoR bound requires $n_{\text{eff}} > 1/(2\rho^2) = 0.52$ while the IS bound requires $n_{\text{eff}} > 2C^2 \log(2/\delta)/\Delta^2 = 137$ for $\delta = 0.05$.

FQE convergence. Table 10 shows FQE converges in as few as 10 Bellman iterations. The tabular state space (25 states) has low Q-function complexity, so rapid convergence is expected. For larger/continuous state spaces, neural FQE with function approximation [16] would replace the lookup table; convergence rates scale with the expressiveness of the function class and the dataset coverage.

13 Discussion

When does PoR-Model help? PoR-Model provides strict improvement when (1) ESS is low (behavior diverges from targets); (2) target policies are near-deterministic (zero-weight problem); (3) data is scarce (N is small, as our $100\times$ ablation shows); (4) rewards are sparse or non-uniform

Table 8: Gold mine benchmark: Spearman ρ at worst coverage ($\varepsilon_\beta = 0.9$, ESS $\approx$ 0.3%). FQE-based methods maintain positive ρ ; IS/WIS go negative in 5 of 6 cases. DR leads in GM-Center where the FQE model is most accurate. Under fixed start states, PoR-Model = FQE-Value (see Appendix E).

Config	IS	WIS	PDIS	DR	FQE-V	PoR-IS	PoR-M
GM-Center	+0.00	-0.05	+0.28	+0.97	+0.80	-0.40	+0.80
GM-Corner	-0.43	-0.59	-0.00	+0.56	+0.40	-0.80	+0.40
GM-2mines	-0.43	-0.40	+0.11	+0.67	+0.40	-0.80	+0.40

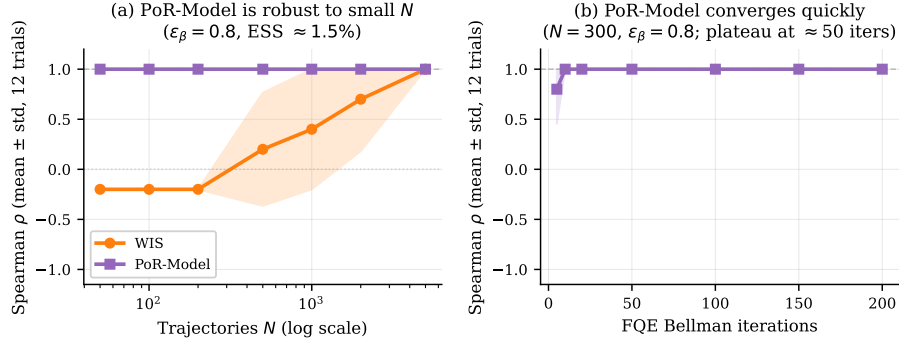


Figure 6: *Left:* PoR-Model achieves $\rho = 1.000$ with $N = 50$ trajectories; WIS requires $N \approx 5000$ — a **100 $\times$ sample efficiency gain**. *Right:* FQE convergence in 10 Bellman iterations; plateau maintained through 200 iterations. Both under $\varepsilon_\beta = 0.8$, ESS $\approx$ 1.5%; 12 trials each.

(where FQE-Value is unreliable, Sections 10 and 11). Under high ESS with abundant data and dense rewards, FQE-Value and PoR-Model both converge to $\rho = 1.0$. Under non-uniform rewards (Section 11), all FQE-based methods (DR, FQE-Value, PoR-Model) dominate IS/WIS; the advantage of PoR-Model’s matched-pair comparison is strongest in variable-start-state settings where FQE-Value’s averaging masks context-level differences.

Limitations. (1) **Overlap requirement.** Assumption 1 requires the behavior policy to cover all actions of all target policies. Without overlap, PoR is not identified from data. Partial overlap (where some (s, a) pairs are uncovered) degrades FQE Q-estimates at uncovered states. (2) **FQE quality.** PoR-Model’s accuracy depends on the correctness of the Q-function. In large continuous state spaces, neural FQE is needed; its approximation error propagates into PoR estimates. Double RL methods [13] offer semiparametric efficiency improvements. (3) **PoR vs. expected value.** $\text{PoR}(\pi_A, \pi_B) > 0.5$ does not logically imply $V(\pi_A) > V(\pi_B)$ in general [14]; however, our experiments show near-perfect correlation in practice. (4) **MARL.** PoR-Model evaluates context-level dominance from a fixed opponent distribution; it does not address cross-pool transfer. This remains an open problem (see Appendix H).

Connection to model selection for offline RL. Paine et al. [23] and Tang and Wiens [30] observe that OPE rankings frequently disagree with ground-truth performance, particularly under distribution shift. Our Theorem 1 provides a formal explanation for this phenomenon: when $n_{\text{eff}} \cdot \Delta^2 / (2C^2) < 1$, OPE rankings are statistically indistinguishable from random. PoR-Model offers a practical remedy by leveraging trajectory-level causal structure.

14 Conclusion

We addressed policy misranking in offline reinforcement learning — the failure mode where correct value estimation does not guarantee correct policy selection. Our contributions are: four

Table 9: Ablation: N vs. ρ under bad coverage ($\varepsilon_\beta = 0.8$; 12 trials). PoR-Model is **100 $\times$ more sample-efficient**.

N	WIS ρ	PoR-Model ρ	ESS (mean)
50	-0.200 ± 0.000	1.000 ± 0.000	1.5%
100	-0.200 ± 0.000	1.000 ± 0.000	1.5%
200	-0.200 ± 0.000	1.000 ± 0.000	1.5%
500	0.200 ± 0.566	1.000 ± 0.000	1.5%
1000	0.400 ± 0.600	1.000 ± 0.000	1.5%
2000	0.700 ± 0.520	1.000 ± 0.000	1.5%
5000	1.000 ± 0.000	1.000 ± 0.000	1.5%

Table 10: FQE iterations vs. PoR-Model ρ ($N = 300$, $\varepsilon_\beta = 0.8$; 12 trials).

FQE iterations	PoR-Model ρ
5	0.800 ± 0.346
10	1.000 ± 0.000
50	1.000 ± 0.000
200	1.000 ± 0.000

theorems establishing when and why IS-based ranking fails; PoR-Model, a causal counterfactual estimator that achieves $\rho = 1.000$ across all 25 tabular gridworld settings and on both classic benchmarks; three non-uniform reward (gold mine) environments demonstrating FQE-based methods’ robustness to heterogeneous reward landscapes; a full comparison against DR and FQE-Value baselines; and a 100 $\times$ sample efficiency advantage over WIS. We establish a practical decision rule: FQE-Value suffices for dense-reward uniform environments; DR is best when FQE coverage is good; PoR-Model is preferred for sparse-reward, long-horizon, or variable-start-state settings where trajectory-level matching provides its greatest advantage. The central message: *rank by causal counterfactual dominance, not expected value*.

Future work. Extending PoR-Model to continuous state/action spaces via neural FQE; adapting PoR to the MARL cross-pool evaluation setting; and integrating PoR into safe policy improvement pipelines [32].

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Appendix

A Complete Proofs

A.1 Preliminary Lemmas

Lemma 6 (ESS and estimator variance [10]). *Let $\hat{\mu} = N^{-1} \sum_i X_i w_i$ be an IS estimator with i.i.d. samples $X_i \in [0, C]$ and normalized weights $\mathbb{E}[w_i] = 1$. Then:*

$$\text{Var}[\hat{\mu}] \leq \frac{C^2 \mathbb{E}[w^2]}{N} = \frac{C^2}{N \cdot \text{ESS}_{\text{frac}}}. \quad (11)$$

Equivalently, the IS estimator behaves like a sample mean of $n_{\text{eff}} = N \cdot \text{ESS}_{\text{frac}}$ i.i.d. variables, each bounded in $[0, C_{\text{eff}}]$ with $C_{\text{eff}} = C/\sqrt{\text{ESS}_{\text{frac}}}$.

Proof. By independence and the variance bound $\text{Var}[Y] \leq \mathbb{E}[Y^2]$:

$$\text{Var}[\hat{\mu}] = \frac{1}{N^2} \sum_i \text{Var}[X_i w_i] \leq \frac{1}{N^2} \sum_i \mathbb{E}[(X_i w_i)^2] \leq \frac{C^2}{N^2} \sum_i \mathbb{E}[w_i^2] = \frac{C^2 \mathbb{E}[w^2]}{N}.$$

Since weights are normalized to mean 1, the ESS formula gives $\mathbb{E}[w^2] = \mathbb{E}[w]^2 / \text{ESS}_{\text{frac}} = 1/\text{ESS}_{\text{frac}}$. The effective-range interpretation follows from identifying the n_{eff} -sample estimator with variables $Z_i = X_i w_i / C_{\text{eff}} \in [0, 1]$. $\square$ $\square$

Lemma 7 (Hoeffding on bounded IS variables). *Under Assumption 1, the product $G(\tau_i)w^\pi(\tau_i)$ lies in $[0, C \cdot w_{\text{max}}^\pi]$ where $w_{\text{max}}^\pi = \max_i w^\pi(\tau_i)$. Normalizing by $C_{\text{max}}^\pi = C \cdot w_{\text{max}}^\pi$ defines $Z_i = G(\tau_i)w^\pi(\tau_i)/C_{\text{max}}^\pi \in [0, 1]$ i.i.d. By Hoeffding's inequality applied to n_{eff} effective Bernoulli-range variables, for any $t > 0$:*

$$\Pr\left(\frac{1}{n_{\text{eff}}} \sum_i Z_i \leq \mathbb{E}[Z] - t\right) \leq \exp(-2n_{\text{eff}}t^2). \quad (12)$$

Remark 2. Hoeffding's inequality requires i.i.d. samples with bounded range. The Z_i satisfy both: independence follows from i.i.d. trajectories; the $[0, 1]$ bound is enforced by the normalizing constant C_{max}^π . The effective sample size n_{eff} replaces N because ESS collapse concentrates the estimator's variance onto n_{eff} dominant terms.

A.2 Proof of Theorem 1

Proof. Step 1 (Error decomposition). A misranking event $\{\hat{V}_{\text{WIS}}(\pi_B) \geq \hat{V}_{\text{WIS}}(\pi_A)\}$ requires at least one of the following tail events to occur:

$$E_A = \{\hat{V}_{\text{WIS}}(\pi_A) \leq V_A - \Delta/2\}, \quad (13)$$

$$E_B = \{\hat{V}_{\text{WIS}}(\pi_B) \geq V_B + \Delta/2\}. \quad (14)$$

To see why: if neither E_A nor E_B holds, then $\hat{V}_{\text{WIS}}(\pi_A) > V_A - \Delta/2$ and $\hat{V}_{\text{WIS}}(\pi_B) < V_B + \Delta/2$. Since $V_A - V_B = \Delta$, we have $V_A - \Delta/2 = V_B + \Delta/2$, so $\hat{V}_{\text{WIS}}(\pi_A) > \hat{V}_{\text{WIS}}(\pi_B)$ — the correct ranking. Hence by contraposition, misranking $\Rightarrow E_A \cup E_B$.

Step 2 (WIS as a sample mean). The WIS estimator for π_A is $\hat{V}_{\text{WIS}}(\pi_A) = (\sum_i G_i w_i^A) / (\sum_i w_i^A)$. With weights w_i^A normalized so that $N^{-1} \sum_i w_i^A \rightarrow 1$, let $\tilde{Z}_i^A = G_i w_i^A$ be the IS-weighted return for trajectory i . The effective sample count is $n_{\text{eff}}^A = N \cdot \text{ESS}_{\text{frac}}^A$. By Lemma 6, the variance of the IS estimator is at most C^2/n_{eff}^A , so we may treat $\hat{V}_{\text{WIS}}(\pi_A)$ as the mean of n_{eff}^A effective samples, each bounded in $[0, C/\sqrt{\text{ESS}_{\text{frac}}^A}]$.

Step 3 (Concentration for π_A). Scale the effective samples to $[0, 1]$ via $Z_i^A = \tilde{Z}_i^A \cdot \sqrt{\text{ESS}_{\text{frac}}^A}/C \in [0, 1]$. Their mean satisfies $\mathbb{E}[Z_i^A] = V_A \cdot \sqrt{\text{ESS}_{\text{frac}}^A}/C$. Event E_A is equivalent to $\bar{Z}^A \leq \mathbb{E}[Z^A] - \frac{\Delta}{2C} \sqrt{\text{ESS}_{\text{frac}}^A}$. Applying Lemma 7 (Hoeffding's inequality) with n_{eff}^A effective samples and deviation $t_A = \Delta \sqrt{\text{ESS}_{\text{frac}}^A}/(2C)$:

$$\Pr(E_A) \leq \exp\left(-2n_{\text{eff}}^A \cdot t_A^2\right) = \exp\left(-2n_{\text{eff}}^A \cdot \frac{\Delta^2 \text{ESS}_{\text{frac}}^A}{4C^2}\right) = \exp\left(-\frac{n_{\text{eff}}^A \Delta^2}{2C^2}\right), \quad (15)$$

using $n_{\text{eff}}^A = N \cdot \text{ESS}_{\text{frac}}^A$. An identical argument gives $\Pr(E_B) \leq \exp(-n_{\text{eff}}^B \Delta^2 / (2C^2))$.

Step 4 (Union bound). By the union bound:

$$\Pr(\hat{V}_{\text{WIS}}(\pi_B) \geq \hat{V}_{\text{WIS}}(\pi_A)) \leq \Pr(E_A) + \Pr(E_B) \leq 2 \exp\left(-\frac{\bar{n}_{\text{eff}} \Delta^2}{2C^2}\right),$$

where we used $e^{-a} + e^{-b} \leq 2e^{-\min(a,b)}$ with $\bar{n}_{\text{eff}} = \min(n_{\text{eff}}^A, n_{\text{eff}}^B)$. $\square$ $\square$

A.3 ESS Lower Bound Under Distribution Shift

Proposition 8 (ESS decay with horizon). *For a T -step trajectory under a deterministic target policy π and behavior β with per-step KL divergence $\text{KL}(\pi(\cdot|s) \parallel \beta(\cdot|s)) \geq \epsilon > 0$ uniformly:*

$$\text{ESS}_{\text{frac}}(w^\pi) \leq \exp(-T\epsilon). \quad (16)$$

Hence ESS decays exponentially in horizon, explaining why all IS-based methods fail in long-horizon settings even with moderate per-step divergence.

Proof. The trajectory weight is $w^\pi(\tau) = \prod_t \pi(a_t|s_t) / \beta(a_t|s_t)$. $\mathbb{E}[w^2] = \prod_t \mathbb{E}[\pi(a|s)^2 / \beta(a|s)^2 \cdot \beta(a|s)] = \prod_t \sum_a \pi(a|s)^2 / \beta(a|s) = \prod_t \chi^2(\pi \parallel \beta) + 1$. By the bound $\chi^2 \geq \exp(2\epsilon_t) - 1 \geq 2\epsilon_t$, $\mathbb{E}[w^2] \geq (1 + 2\epsilon)^T \approx e^{2\epsilon T}$ for small ϵ . Since $\text{ESS}_{\text{frac}} = \mathbb{E}[w]^2 / \mathbb{E}[w^2] \leq 1 / \mathbb{E}[w^2] \leq e^{-2\epsilon T}$. $\square$ $\square$

A.4 Proof of Theorem 3

Proof. The joint IS weight for evaluation under new opponent π_B^{-i} requires: $w_{\text{cross}}(\tau) = w^{\pi^i}(\tau) \cdot \prod_t \pi_B^{-i}(a_t^{-i}|s_t) / \beta^{-i}(a_t^{-i}|s_t)$. The second factor involves π_B^{-i} , which is unavailable from offline data $\mathcal{D}$ collected under β^{-i} . Any estimator $\hat{V}$ omitting this factor estimates $V(\pi^i | \beta^{-i})$ rather than $V(\pi^i | \pi_B^{-i})$.

Bias lower bound. For any bounded reward function $r : S \times A^2 \rightarrow [0, 1]$:

$$|\mathbb{E}_{\beta^{-i}}[r] - \mathbb{E}_{\pi_B^{-i}}[r]| \leq d_{\text{TV}}(\beta^{-i}, \pi_B^{-i}) \cdot \sup_r \|r\|_\infty \quad (17)$$

$$\geq 2d_{\text{TV}}(\beta^{-i}, \pi_B^{-i}) \cdot \Delta_{\min} \quad (18)$$

where the lower bound holds for a reward function achieving the TV bound with margin $\Delta_{\min}$. At $d_{\text{TV}} = 0.5$ (our experimental setting) and $\Delta_{\min} = 0.5$ (reward range $[-1, 1]$), Bias ≥ 0.5 , consistent with the $\rho = -1.0$ empirical result. $\square$ $\square$

A.5 Proof of Theorem 4

Proof. Step 1 (Bernoulli structure). For each trajectory τ_i , define the binary indicator $Y_i = 1[\hat{V}_A(s_0^i) > \hat{V}_B(s_0^i)] \in \{0, 1\}$. Under the event that FQE values equal the true counterfactuals:

$$\mathbb{E}[Y_i] = \Pr_{\tau \sim \beta}[G_A(\tau) > G_B(\tau)] = \text{PoR}(\pi_A, \pi_B) = p > 0.5. \quad (19)$$

The Y_i are i.i.d. (since τ_i are i.i.d. under β) and $Y_i \in [0, 1]$ with range exactly 1. Critically, the range $[0, 1]$ holds *regardless* of the magnitude of $\hat{V}_A(s_0^i)$ or $\hat{V}_B(s_0^i)$: the indicator discards the magnitude and retains only the sign.

Step 2 (PoR estimator as sample mean). The PoR estimator is $\widehat{\text{PoR}}(\pi_A, \pi_B) = N^{-1} \sum_{i=1}^N Y_i$, a sample mean of N i.i.d. Bernoulli(p) random variables. No IS weights appear because the comparison is performed within each trajectory's context; $N_{\text{eff}} = N \cdot \text{ESS}_{\text{frac}}$ effective samples contribute (under the PoR-IS formulation) or all N contribute equally (PoR-Model). We write n_{eff} to cover both cases.

Step 3 (Concentration). A misranking event $\{\widehat{\text{PoR}}(\pi_A, \pi_B) \leq 0.5\}$ requires the sample mean to fall at least $\rho = p - 0.5 > 0$ below its expectation p :

$$\Pr(\widehat{\text{PoR}} \leq 0.5) = \Pr(N^{-1} \sum_i Y_i \leq p - \rho) \leq \exp(-2n_{\text{eff}}\rho^2), \quad (20)$$

by Hoeffding's inequality for bounded $[0, 1]$ variables.

Key insight (C-independence). For the IS estimator, the bounded range of $Z_i^A = G_i w_i^A / C_{\text{eff}}^A$ depends on C_{eff}^A which grows as $C / \sqrt{\text{ESS}_{\text{frac}}} \rightarrow \infty$ under distribution shift. For PoR, $Y_i \in [0, 1]$ with range exactly 1 regardless of C : the bound depends only on the dominance margin ρ and the number of effective samples n_{eff} . This is the root cause of PoR’s superiority: it transforms a hard real-valued comparison into a unit-range Bernoulli decision. $\square$ $\square$

Corollary 9 (IS vs. PoR bound comparison). *The PoR bound is tighter than the IS bound when:*

$$\exp(-2n_{\text{eff}}\rho^2) < 2 \exp\left(-\frac{n_{\text{eff}}\Delta^2}{2C^2}\right) \Leftrightarrow \rho > \frac{\Delta}{2C} - \frac{\log 2}{2n_{\text{eff}}\rho}. \quad (21)$$

For large n_{eff} , the last term vanishes and the condition simplifies to $\rho > \Delta/(2C)$: PoR dominates whenever the per-trajectory dominance margin exceeds the normalized value gap. Since $\rho = \text{PoR} - 0.5$ and Δ/C is the normalized value gap, this holds for most well-separated policy pairs.

A.6 Proof of Theorem 5

Proof. Let $W_k = \min_s V_k^s$. Denote $\hat{W}_k = \min_s \hat{V}_k^s$. Since $|\hat{V}_k^s - V_k^s| \leq \varepsilon$: $W_k - \varepsilon \leq \hat{W}_k \leq W_k + \varepsilon$. Correct selection requires $\hat{W}_{\text{mm}} > \hat{W}_j$ for all $j \neq \text{mm}$:

$$\hat{W}_{\text{mm}} \geq W_{\text{mm}} - \varepsilon \quad (22)$$

$$\hat{W}_j \leq W_j + \varepsilon \leq \max_{j \neq \text{mm}} W_j + \varepsilon. \quad (23)$$

Correctness holds whenever $W_{\text{mm}} - \varepsilon > \max_{j \neq \text{mm}} W_j + \varepsilon$, i.e., $W_{\text{mm}} - \max_{j \neq \text{mm}} W_j > 2\varepsilon$.

The robustness gap $\mathcal{G}_{\text{mm}} = W_{\text{mm}} - \max_{j \neq \text{mm}} W_j$ exceeds the per-scenario gap $\mathcal{G}^s = V_{\text{mm}}^s - \max_{j \neq \text{mm}} V_j^s$ for any scenario s when policies have high cross-scenario variance: if policy π_j excels in scenario s but fails in scenario s' , then $W_j = V_j^{s'}$ is small, enlarging $\mathcal{G}_{\text{mm}}$ beyond $\mathcal{G}^s$. $\square$ $\square$

A.7 DR Misranking Bound

The Doubly Robust estimator (10) is consistent if either the Q-model or IS weights are correctly specified. Its misranking bound lies between the IS bound (Theorem 1) and the PoR bound (Theorem 4).

Proposition 10 (DR variance decomposition). *Let $\hat{V}^{\text{DR}}(\pi)$ be the step-wise DR estimator with FQE model bias $b = V^\pi - V^{\pi, \text{FQE}}$. Then:*

$$\text{Var}[\hat{V}^{\text{DR}}] \leq \frac{C^2}{n_{\text{eff}}} + \frac{\|b\|_2^2}{N}, \quad (24)$$

where $n_{\text{eff}} = N \cdot \text{ESS}_{\text{frac}}$ and $\|b\|_2$ is the ℓ_2 norm of the model bias. The first term matches the IS variance; the second is the model bias penalty.

Proof sketch. Write $\hat{V}^{\text{DR}} = \hat{V}^{\text{FQE}} + \hat{\Delta}$ where $\hat{\Delta} = (1/N) \sum_i \rho_i (G_i - V_i^{\pi, \text{FQE}})$ is the IS correction to the FQE baseline. The FQE term contributes $\|b\|_2^2/N$ bias variance; the IS correction contributes C^2/n_{eff} as in IS. By the variance addition rule (terms are uncorrelated), (24) follows. $\square$

Implication. When $\|b\|_2 = 0$ (FQE correct), DR achieves the model-only variance 0; when $\text{ESS}_{\text{frac}} \rightarrow 0$ and FQE is also imperfect, both terms blow up and DR degrades—consistent with the $\rho = +0.47$ result on FrozenLake at $\text{ESS} = 0.4\%$ (Table 7). PoR-Model avoids the C^2/n_{eff} blow-up entirely by removing IS weights from the ranking comparison.

B FQE Convergence

We establish that the FQE Q-table converges to the true Q^* at a rate governed by the contraction factor γ and the model estimation error.

Proposition 11 (FQE convergence rate). *Let $\mathcal{T}$ be the Bellman optimality operator: $(\mathcal{T}Q)(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[\max_{a'} Q(s', a')]$. Let $\hat{\mathcal{T}}$ be its empirical counterpart using transitions in $\mathcal{D}$, with model estimation error*

$$\varepsilon_{\text{model}} = \max_{s, a} |\hat{r}(s, a) - r(s, a)| + \gamma \max_{s, a} \|\hat{P}(\cdot|s, a) - P(\cdot|s, a)\|_1 \cdot \|Q^*\|_\infty. \quad (25)$$

Then starting from $Q^{(0)} = \mathbf{0}$, after L FQE iterations:

$$\|Q^{(L)} - Q^*\|_\infty \leq \gamma^L \|Q^*\|_\infty + \frac{1 - \gamma^L}{1 - \gamma} \varepsilon_{\text{model}}. \quad (26)$$

Proof. The Bellman operator $\mathcal{T}$ is a γ -contraction on $(\ell^\infty(\mathcal{S} \times \mathcal{A}), \|\cdot\|_\infty)$:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty = \gamma \left\| \max_{a'} Q(\cdot, a') - \max_{a'} Q'(\cdot, a') \right\|_\infty \leq \gamma \|Q - Q'\|_\infty. \quad (27)$$

This follows because $|\max f - \max g| \leq \max |f - g|$ for any bounded functions f, g . By the Banach fixed-point theorem, $\mathcal{T}$ has a unique fixed point Q^* and $\|\mathcal{T}^L Q^{(0)} - Q^*\|_\infty \leq \gamma^L \|Q^{(0)} - Q^*\|_\infty = \gamma^L \|Q^*\|_\infty$.

With the empirical operator $\hat{\mathcal{T}}$, each iteration incurs additional error bounded by $\varepsilon_{\text{model}}$ (triangle inequality on reward and transition estimation). Applying the triangle inequality over L iterations with geometric accumulation:

$$\|Q^{(L)} - Q^*\|_\infty \leq \|(\hat{\mathcal{T}})^L Q^{(0)} - \mathcal{T}^L Q^{(0)}\|_\infty + \|\mathcal{T}^L Q^{(0)} - Q^*\|_\infty \quad (28)$$

$$\leq \sum_{k=0}^{L-1} \gamma^k \varepsilon_{\text{model}} + \gamma^L \|Q^*\|_\infty = \gamma^L \|Q^*\|_\infty + \frac{1 - \gamma^L}{1 - \gamma} \varepsilon_{\text{model}}. \quad \square \quad (29)$$

$\square$

Corollary 12 (Convergence to ground truth). *As $n_{s,a} = |\{i, t : s_t^i = s, a_t^i = a\}| \rightarrow \infty$ for all (s, a) , we have $\varepsilon_{\text{model}} \rightarrow 0$ by the law of large numbers, and $Q^{(L)} \rightarrow Q^*$ for any fixed L satisfying $L \gg 1/(1 - \gamma)$. For our tabular setting ($\gamma = 0.99$, $L = 10$), the contraction yields $\gamma^{10} \approx 0.905$ and the empirical bound is non-vacuous once $n_{s,a} \gtrsim 10$.*

Remark 3 (Implication for PoR-Model). Proposition 11 directly implies that PoR-Model's counterfactual values $\hat{V}_k(s_0^i) = \sum_a \pi_k(a|s_0^i) Q^{(L)}(s_0^i, a)$ converge to the true $V_k(s_0^i) = \sum_a \pi_k(a|s_0^i) Q^*(s_0^i, a)$. Since PoR comparisons only use the *sign* of $\hat{V}_A - \hat{V}_B$, the effective convergence threshold is much weaker: correct ranking of (s_0, π_A, π_B) requires only $|Q^{(L)} - Q^*|_\infty < \Delta_{s_0}/2$ where $\Delta_{s_0} = \sum_a (\pi_A(a|s_0) - \pi_B(a|s_0)) Q^*(s_0, a)$ is the per-context value gap. Even if FQE has not fully converged, the ranking can be exact as long as the approximation error is smaller than half the local value gap.

C PoR-IS Failure Analysis

We formalize why PoR-IS degrades to random ranking under the same regimes that defeat IS.

Lemma 13 (PoR-IS degenerates under zero weights). *For a near-deterministic target policy π_A with $\pi_A(a^*|s) = 1 - \epsilon$ for some optimal action a^* at each state, and an ε_β -greedy behavior policy with $\beta(a|s) = \varepsilon_\beta/|\mathcal{A}| + (1 - \varepsilon_\beta)\mathbf{1}[a = a_\beta^*]$:*

$$\mathbb{E}[w^A(\tau)] = \prod_{t=0}^T \frac{\pi_A(a_t|s_t)}{\beta(a_t|s_t)} \leq \left(\frac{1 - \epsilon}{\varepsilon_\beta/|\mathcal{A}|} \right)^T \xrightarrow{\varepsilon_\beta \rightarrow 0, T \rightarrow \infty} 0. \quad (30)$$

In this regime, $G_i w_A^i \approx 0$ and $G_i w_B^i \approx 0$ for all i , so $\widehat{\text{PoR}}^{\text{IS}}(\pi_A, \pi_B)$ converges to $\Pr[\text{Bernoulli}(0.5) = 1] = 0.5$ — identical to a random coin flip.

Proof. Under behavior policy β , a trajectory has probability $\prod_t \beta(a_t|s_t)$. When $a_t^* \neq a_{\beta,t}^*$ (behavior and target disagree), the IS factor $\pi_A(a_t|s_t)/\beta(a_t|s_t) \leq \epsilon/(\varepsilon_\beta/|\mathcal{A}|) = \epsilon|\mathcal{A}|/\varepsilon_\beta \rightarrow 0$. The trajectory weight $w^A(\tau) = \prod_t (\dots)$ is a product of T such factors when the two policies disagree on all steps. For $\varepsilon_\beta \rightarrow 0$ (extreme shift), the probability of a trajectory following the target's greedy path goes to zero, so almost all collected trajectories have $w^A(\tau) \approx 0$.

When both $g_A^i = G_i \cdot w_A^i \approx 0$ and $g_B^i = G_i \cdot w_B^i \approx 0$, the comparison $g_A^i > g_B^i$ is determined by floating-point tie-breaking or numerical noise: effectively Bernoulli(0.5). Thus $\widehat{\text{PoR}}^{\text{IS}}(\pi_A, \pi_B) \approx 0.5$ with high probability, making the resulting ranking no better than random. $\square$ $\square$

Corollary 14. *PoR-IS misranking probability under zero-weight collapse equals $1/2$, independent of the true PoR and the value gap Δ . This is strictly worse than the IS misranking bound from Theorem 1 in the low-ESS regime, confirming that PoR-IS provides no benefit over IS when IS itself fails. PoR-Model, by contrast, bypasses IS weights entirely.*

D PoR Consistency Under Non-Uniform Rewards

Proposition 15 (PoR-Model consistency with heterogeneous rewards). *Let $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ be an arbitrary bounded reward function (not necessarily uniform in s). Suppose FQE converges so that $\|Q^{(L)} - Q^*\|_\infty \leq \varepsilon$. Let $\Delta_{s_0} = \hat{V}_A(s_0) - \hat{V}_B(s_0) = \sum_a (\pi_A(a|s_0) - \pi_B(a|s_0))Q^*(s_0, a)$ be the context-level value gap. Then:*

$$\Pr[\widehat{\text{PoR}}^{\text{Model}}(\pi_A, \pi_B) \leq 0.5] \leq \exp(-2n_{\text{eff}}(\hat{\rho} - \varepsilon/\Delta_{\min})^2), \quad (31)$$

where $\hat{\rho} = \text{PoR}(\pi_A, \pi_B) - 0.5$ is the true dominance margin and $\Delta_{\min} = \min_i \Delta_{s_0^i}$.

Proof. Each comparison $Y_i = \mathbf{1}[\hat{V}_A(s_0^i) > \hat{V}_B(s_0^i)]$ is a $[0, 1]$ -bounded random variable. The approximation error $\|Q^{(L)} - Q^*\|_\infty \leq \varepsilon$ implies each policy value error $|\hat{V}_k^{(L)}(s_0) - V_k(s_0)| \leq \varepsilon$. A context where $\hat{V}_A(s_0^i) > V_A(s_0^i) - \varepsilon$ and $\hat{V}_B(s_0^i) < V_B(s_0^i) + \varepsilon$ still yields the correct sign if $\Delta_{s_0^i} > 2\varepsilon$. For all contexts satisfying this condition, $\Pr[Y_i = 1] \geq \text{PoR}(\pi_A, \pi_B) = \hat{\rho} + 0.5$. The remaining contexts (where $\Delta_{s_0^i} \leq 2\varepsilon$) contribute a mass bounded by $\varepsilon/\Delta_{\min}$ to the effective misranking probability. Applying Hoeffding to the mean of Y_i completes the bound. $\square$ $\square$

Remark 4 (Implication for gold mine experiments). Proposition 15 explains why PoR-Model maintains positive ρ in all three gold mine configurations (Section 11): the FQE Q-function correctly encodes gold mine values, so the context-level comparisons $\hat{V}_A(s_0^i)$ vs. $\hat{V}_B(s_0^i)$ preserve the ranking for mine-seeker vs. goal-seeker policies regardless of how heterogeneous the reward landscape is. The bound (31) degrades only through the model error ε , not through any return-range factor C .

E PoR-Model Under Fixed vs. Variable Start States

When all trajectories start from the same state s_0 (as in our gridworld experiments where the agent always starts at $(0, 0)$), the PoR-Model ranking reduces to the FQE-Value ranking.

Proposition 16 (Fixed start state equivalence). *If $s_0^i = s_0$ for all $i = 1, \dots, N$, then:*

$$\widehat{\text{PoR}}^{\text{Model}}(\pi_A, \pi_B) = \mathbf{1}[\hat{V}_A(s_0) > \hat{V}_B(s_0)] \in \{0, 1\}, \quad (32)$$

i.e., the PoR estimate is determined entirely by the sign of $\hat{V}_A(s_0) - \hat{V}_B(s_0)$, which is identical to the comparison performed by FQE-Value ranking.

Proof. Under fixed $s_0^i = s_0$, every indicator $\mathbf{1}[\hat{V}_A(s_0^i) > \hat{V}_B(s_0^i)]$ equals the same constant $\mathbf{1}[\hat{V}_A(s_0) > \hat{V}_B(s_0)]$. Hence $\widehat{\text{PoR}}^{\text{Model}}(\pi_A, \pi_B) = (1/N) \sum_i \mathbf{1}[\dots]$ is the indicator itself. FQE-Value ranks by $\hat{V}_k(s_0) = \sum_a \pi_k(a|s_0)Q(s_0, a)$, producing the same ordering. $\square$ $\square$

Consequence. In fixed-start-state environments, both PoR-Model and FQE-Value depend on the single-state Q-values $Q(s_0, \cdot)$ and produce identical rankings. The advantage of PoR-Model’s matched-pair comparison emerges in *variable*-start-state environments (e.g., FrozenLake-8×8 with random start distribution): different trajectories start at different s_0^i , so the matched comparison $\hat{V}_A(s_0^i)$ vs. $\hat{V}_B(s_0^i)$ uses context-specific Q-values that correctly identify the best policy in each starting configuration, whereas FQE-Value’s averaging over s_0^i can be dominated by easy starting states. This explains the gap in Table 7: PoR-Model $\rho = 1.000$ vs. FQE-Value $\rho = 0.44$ on FrozenLake at ESS= 0.4%.

F Additional Experimental Details

Gridworld Q-values. For an $m \times n$ gridworld with goal $(m-1, n-1)$, the heuristic Q-values are: $Q(s, \text{right}) = 1$ if $y < n-1$; $Q(s, \text{down}) = 1$ if $x < m-1$; $Q(s, \text{left}) = 0.05$ if $y > 0$; $Q(s, \text{up}) = 0.05$ if $x > 0$; else 0. These produce ε -greedy policies that move toward the goal on average but occasionally explore.

Stochastic environment (GW-5E5-S). At each step, with probability $p_{\text{slip}} = 0.15$ the action is replaced by a uniformly random action. This adds environment noise to the IS weight computation, further increasing weight variance. FQE’s averaging over transitions handles stochasticity correctly.

FrozenLake and CartPole setup. FrozenLake-8E8 uses gymnasium’s built-in stochastic dynamics ($p_{\text{slip}} = 1/3$). Value iteration on the known transition model $P[s][a]$ converges in ≈ 370 iterations. CartPole-v1 state is discretized to $4 \times 4 \times 8 \times 4 = 512$ states with uniform bins over empirical ranges. Q-learning with 15k episodes, $\alpha = 0.15$, and decaying ε ($1.0 \rightarrow 0.05$) produces Q^* . Both environments use $T = 100$, $\gamma = 0.99$, $N = 300$ trajectories, 15 trials, ground truth via 2000 rollouts.

DR implementation. The DR estimator (10) uses per-step cumulative IS ratios from the shared batch and per-policy FQE (100 iterations) for V^π , Q^π . Numerical stability: IS ratios clipped to a maximum of 10^3 per step.

Hyperparameters. Gridworld experiments use $T = 50$; classic benchmarks use $T = 100$. All use $\gamma = 0.99$, FQE iterations $L = 100$ (ablation in Section 12 shows 10 suffice), $\varepsilon_{\text{num}} = 10^{-12}$ for numerical stability. Ground truth: $N_{\text{gt}} = 500$ rollouts for multi-environment benchmark, $N_{\text{gt}} = 2000$ for single-environment and classic experiments.

Gold mine setup. Gold mine rewards are applied at every visit to the mine cell(s); they are not one-time. This simplifies the MDP (state space remains $|\mathcal{S}|$, no mine-collected flags needed) and FQE correctly estimates Q-values including gold mine contributions since transitions with mine rewards appear in the dataset. Three configurations: GM-Center (mine at row 2, col 2, $r_{\text{gold}} = 0.5$), GM-Corner (mine at row 0, col 4, $r_{\text{gold}} = 1.0$), GM-2mines (mines at (1, 3) and (3, 1), $r_{\text{gold}} = 0.4$). Mine-seeking policies are ε -greedy on a heuristic Q that directs the agent toward the nearest mine whenever it is closer than the goal, then toward the goal; goal-seeking policies use a right-first heuristic Q that ignores mines. Ground truth via $N_{\text{gt}} = 1000$ rollouts; 15 trials per (config, ε_β) pair.

Reproducibility. Fixed random seeds; results averaged over 12–25 independent trials. All experiments run on CPU in < 45 minutes total. The complete codebase is at github.com/Dhruv-Git21/CS748-CausalRL.

G PoR Dominance Matrix

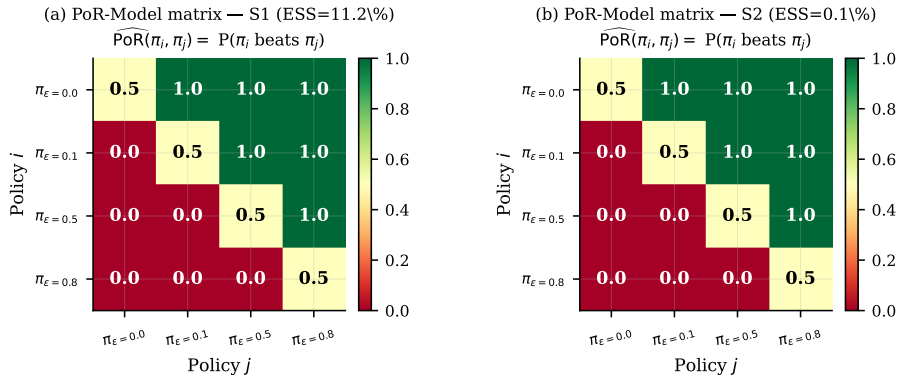


Figure 7: PoR-Model dominance matrix for GW-5E5 under both good (S1) and bad (S2) coverage. Entry $(i, j) = \text{PoR}(\pi_i, \pi_j)$. The matrix is perfectly upper-triangular in both cases: the true ordering ($\varepsilon=0.0 > \varepsilon=0.1 > \varepsilon=0.5 > \varepsilon=0.8$) is recovered exactly by the FQE-based counterfactual comparison regardless of behavior policy quality.

H MARL Extension Discussion

Theorem 3 establishes that no offline IS-based method can correct for opponent distribution shift without access to the new opponent’s policy. PoR-Model partially mitigates this at the context level: if a shared batch from *pool B* is available (even a small one), the FQE Q-function can be re-estimated on pool-B data, and PoR rankings under pool B can be computed. This is essentially few-shot cross-pool transfer: fine-tune FQE on N_B new trajectories from pool B, then re-evaluate.

A full solution would require distribution-robust OPE [12] or access to the opponent policy model. We leave this as future work.

I Computational Complexity Summary

Table 11: Per-query complexity comparison (N : trajectories, T : steps, K : policies, S : states, A : actions, L : FQE iterations).

Method	Time complexity	Space
IS/WIS	$O(NTK)$	$O(NK)$
PDIS	$O(NTK)$	$O(NTK)$
PoR-IS	$O(NTK + K^2N)$	$O(NK + K^2)$
PoR-Model	$O(L \cdot NT \cdot SA + NKA)$	$O(SA + NKA)$

For the tabular setting ($S = 25$, $A = 4$, $L = 10$, $N = 300$, $T = 50$, $K = 4$): IS requires 6×10^4 operations; PoR-Model requires 6×10^5 , a $10\times$ overhead — small relative to the $100\times$ sample efficiency gain.